MONASH University

Developing a Power BI Dashboard for AHRI's Call Activity

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GitHub repository

https://github.com/apra0095/AHRI-Call-Dashboard-Project

Report for
Australian HR Institute (AHRI)

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Table of contents

Abstract **4**

1 Background and objectives **5**
1.1 The organisation and the reporting problem . . . 5
1.2 How calls move through the system . . . 6
1.3 Objectives and contribution . . . 6

2 Data sources and initial data analysis **7**
2.1 Files, coverage and analytical populations . . . 7
2.2 Understanding the main fields . . . 8
2.3 Missing values and their treatment . . . 9
2.4 Duplicate checks and consistent data types . . . 9

3 Analytical approach and Power BI implementation **10**
3.1 Separating data preparation from calculations . . . 10
3.2 Identifying Main IVR entries and related records . . . 10
3.3 Classifying queue destinations . . . 11
3.4 Selecting and counting outbound activity . . . 12
3.5 Time conversion and average activity . . . 12
3.6 Technical outcomes and duration . . . 13
3.7 Relationships and reproducibility . . . 13
3.8 Dashboard navigation and use . . . 14

4 Inbound findings **14**
4.1 Overall demand and the comparison period . . . 14
4.2 Daily variation and unusually busy periods . . . 15
4.3 Changes across fully covered months . . . 15
4.4 Differences between days of the week . . . 16
4.5 When during the day calls arrive . . . 17
4.6 Combining weekday and hour . . . 18
4.7 Recorded queue destinations . . . 18
4.8 Implications for coverage and service review . . . 20

5 Outbound findings **20**
5.1 Connecting incoming demand with staff calling activity . . . 20
5.2 Activity within the approved staff group . . . 21
5.3 Timing and monthly changes . . . 22
5.4 Technical outcomes and recorded duration . . . 23
5.5 What the combined analysis provides . . . 24

6 Statistical summaries and variability **24**
6.1 Why averages need context . . . 25
6.2 Overall activity and duration . . . 25
6.3 How activity varies within each weekday . . . 26
6.4 Variation across fully covered months . . . 26
6.5 What these summaries can establish . . . 27

7 Conclusion **27**

References **29**

Appendix A: Power BI dashboard **30**

Abstract

This project developed a Power BI dashboard to help the Australian HR Institute (AHRI) understand incoming telephone demand and outbound calling activity. Microsoft Teams call exports covering 6 April to 1 September 2026 were examined to establish their structure, quality and reporting scope. The inbound analysis focused on 3,986 linked groups containing a record for AHRI's automated main telephone menu (Main IVR). The outbound analysis covered 6,046 attempts by 18 staff approved by AHRI.

Data preparation included checking missing values and duplicates, standardising account information, converting timestamps to Melbourne time and linking related records. The dashboard presents activity by date, weekday, hour, service and team. Additional analysis in R used means, medians, standard deviations and boxplots to examine variation beyond the averages.

Inbound activity averaged 37.4 entries per weekday, with a standard deviation of 21.7, indicating substantial variation between dates. Tuesday had the highest overall weekday average, and demand remained substantial through lunchtime. Average weekday entries decreased from 53.0 in May to 29.5 in August. Membership accounted for 87.2% of selected outbound attempts, while outbound activity followed a different monthly pattern from inbound demand.

These findings provide a basis for reviewing telephone coverage and calling schedules. However, the exports do not establish waiting times, abandonment or enquiry resolution, and technical success does not confirm a conversation. The project delivers a reproducible dashboard and supporting analysis that help AHRI interpret recorded telephone activity within these limitations.

1 Background and objectives

1.1 The organisation and the reporting problem

The Australian HR Institute (AHRI) is a professional membership body supporting human resources practitioners. Its activities include professional certification, training, practical resources and events. These services generate different reasons for members and prospective members to contact the organisation, from membership enquiries to questions about professional development (Australian HR Institute, n.d.). Phone activity is therefore part of service delivery, alongside email and online enquiries. Understanding this activity matters because the same staff may need to manage several channels during a working day.

At the first meeting, AHRI described a gap between its digital and telephone reporting. Email and online enquiries fed into its Dynamics customer relationship management system, where existing dashboards provided visibility. Telephone activity was handled through Microsoft Teams, but the Member Engagement Team relied on raw exports obtained by IT. Staff could inspect individual records, yet there was no convenient shared view of demand, destinations or calling patterns. The project addressed that reporting gap rather than proposing a replacement telephone system (AHRI and Monash University project team, 2026).

The initial brief asked for call volumes, routing destinations, answer and abandonment rates, waiting time and talk time. It also identified outbound activity as a possible extension. However, a requested metric is not automatically a measurable one. The files eventually available contained technical call records rather than a complete record of queue events or customer conversations. The project therefore had to establish what each row represented before selecting indicators. This distinction became the central analytical challenge, because incorrect counting could produce a visually attractive but misleading dashboard.

The intended audience is AHRI's Member Engagement Team and colleagues responsible for understanding telephone operations. They need a report that supports questions such as when inbound demand is concentrated, which services appear in queue records, and how outbound activity varies across teams. They also need clear boundaries around interpretation. A technical connection status is useful information, but it cannot establish whether a member spoke to an employee or had their enquiry resolved. The dashboard is designed to support investigation and planning without presenting unsupported service-performance claims.

1.2 How calls move through the system

An inbound caller may contact AHRI's main telephone number and hear an automated menu, commonly called an interactive voice response system, or IVR. The caller's choice can direct the call towards a service such as Membership or Events. A call queue is a system component that distributes calls to staff according to its configuration. An overflow destination is an alternative destination configured for particular queue conditions. Microsoft distinguishes queue overflow, timeout and other exception settings. The export alone does not reveal which configured condition actually operated for a particular record (Microsoft, 2026c).

One telephone interaction can produce several records as it moves between system accounts. For example, a person may enter the main menu and then reach a queue. The export can contain a record associated with the menu and another associated with the queue. Counting both as separate incoming customer enquiries would inflate demand. Conversely, counting only rows labelled Main IVR when examining routing would discard related records that help describe the destinations observed. The project uses separate record-level and group-level views to address these different questions.

There is also a route that bypasses the main menu: someone may call a staff member directly. During the project, Vanessa confirmed that calls without a Main IVR record were mostly direct calls to staff. This clarification explains why the export contains more linked groups than Main IVR entries. It does not prove that every remaining group has the same explanation. The report consequently uses the observed label "without a Main IVR record" when discussing the full reconciliation, while using the stakeholder explanation to justify the narrower inbound focus.

1.3 Objectives and contribution

The first objective was to establish defensible counting rules for inbound demand and outbound activity. This required distinguishing source rows, related-record groups, Main IVR entries and outbound attempts. The second objective was to prepare a consistent model that could support date, service, team and staff comparisons. The third was to build clear Power BI pages through which users could explore those comparisons. The fourth was to evaluate the reliability and limits of the resulting measures, including the effects of incomplete dates, missing fields and ambiguous technical outcomes.

The contribution is a practical analytical tool supported by documented definitions and checks. It combines Power Query preparation, a structured semantic model, DAX calcula-

tions and interactive report pages. The value comes from making existing records understandable and reusable rather than developing a new statistical algorithm. A particularly useful contribution is the reconciliation of apparently conflicting call totals. Explaining why several thousand source rows reduce to a smaller number of relevant groups gives users a sound basis for interpreting every subsequent chart.

The scope is retrospective visibility over the supplied export period. Real-time monitoring, service-level targets, customer satisfaction, staff productivity and causal evaluation are outside the current scope. Forecasting was also excluded: the immediate priority was reliable measurement, and the available period is too limited to justify strong claims about recurring annual patterns. This report evaluates the repository snapshot dated 6 October 2026. It reports verified results alongside implementation limitations, and separates proposed improvements from work already completed.

2 Data sources and initial data analysis

2.1 Files, coverage and analytical populations

The project uses two Excel files supplied for AHRI's Microsoft Teams Direct Routing activity. The inbound workbook, `DirectRouting_Ehsan_2026-09-01.xlsx`, contains 9,859 inbound records. The later workbook, `Export_Outbound.xlsx`, contains 17,655 records covering both directions: 9,859 inbound and 7,796 outbound. Despite its filename, the second workbook is not restricted to outbound calls. Filtering Call Direction is therefore necessary before analysing outbound activity. Comparing the common columns also confirms that its inbound records match those in the original workbook.

The observed overall coverage runs from 6 April to 1 September 2026 in Melbourne time. These dates indicate the earliest and latest observed activity, but do not confirm that both boundary days were fully captured. For example, an export could end partway through its final day. The model therefore excludes the first and last observed dates when calculating daily averages. It treats May through August as fully covered months, assuming that the export contains continuous coverage between its boundaries. This assumption needs confirmation when another export is added.

The main reporting populations are summarised in **?@tbl-populations**. These counts describe different levels and subsets of the data, so they should not be added together.

Table 1: *Source populations and counting units, recalculated from the workbooks stored in the project repository.*

Reporting population	Count	Meaning
Original inbound records	9,859	Individual system records
Distinct inbound shared-ID groups	6,361	Groups of related records
Groups containing Main IVR	3,986	Main-menu population used for inbound analysis
Other inbound groups	2,375	Groups without a Main IVR record
All outbound records	7,796	Outbound records before staff selection
Selected outbound attempts	6,046	Attempts associated with the approved staff

The outbound analysis follows the stakeholder's list of staff in Membership, HRSC and Events. It includes eleven Membership staff, five HRSC staff and two Events staff. Their account-to-team mapping is stored in `DimStaff`, using the user principal name, or UPN, as the matching key. Name variations involving Lara and Georgina were clarified during the project before finalising the mapping. Matching against the approved accounts excludes 1,750 outbound records outside the agreed staff list. These records are outside the scope of this analysis, their exclusion does not mean they are incorrect.

2.2 Understanding the main fields

The two main identifiers are `CorrelationId` and `Shared correlation ID`. CorrelationId is a technical identifier associated with a recorded call. Shared correlation ID connects related call records. Neither identifies a unique person. The UPN identifies the associated user or resource account, while Display Name provides a readable account label. Call Direction distinguishes inbound from outbound activity. Invite time records the initial call invitation and provides a timestamp for analysing attempts, including unsuccessful ones. Duration, Success, SIP codes and SIP phrases describe recorded technical activity (Microsoft, 2025, 2026a).

These fields need to be interpreted at the correct level. A shared-ID group can contain several rows associated with different accounts or destinations. It should not automatically

be treated as a complete customer journey, because the export may not contain every event needed to reconstruct that journey. A telephone number also cannot establish the identity of a unique customer. Numbers may be shared, withheld or unavailable. The report therefore avoids presenting shared-ID groups as unique people or using these records to make claims about individual customer behaviour.

2.3 Missing values and their treatment

Initial data analysis checked the number of records, call directions, missing values, duplicate rows, repeated identifiers and timestamp coverage. The inbound workbook contains 334 missing UPN values and 334 missing display names. It also contains 785 records without start and end timestamps. Records with missing account information were retained, with their account identity treated as unknown. Other fields in these rows can still contribute to the analysis, including shared identifiers, invitation timestamps and technical outcomes. Removing the rows would change the reporting population without establishing that the records were invalid.

Missing timestamps were also retained rather than replaced with estimated values. In the inbound workbook, all 785 records without start timestamps have an unsuccessful status. In the selected outbound population, all 542 unsuccessful attempts have missing start timestamps and zero recorded duration. These patterns show why missing values should be considered alongside other fields. Replacing a missing start time with Invite time could imply that a connection occurred when the source does not establish one. Deleting unsuccessful attempts would also understate activity and remove information needed to describe technical outcomes.

The preparation therefore keeps these records while applying the relevant population filter to each measure. Outbound attempt counts include both technically successful and unsuccessful records. Successful-duration measures use records marked successful. Missing account details remain a limitation when identifying destinations, but do not automatically make every other field in the same row unusable. This approach preserves the available evidence without filling gaps with unsupported assumptions.

2.4 Duplicate checks and consistent data types

No exact duplicate rows were identified in the inbound workbook. However, one correlation identifier appears more than once. This distinction matters because a repeated identifier does not necessarily mean that the entire row is duplicated. Records must be compared before deciding whether anything should be removed. Two inbound columns contain no

observed values: an unnamed Excel column and Administrative Units. These columns are excluded during preparation because they contribute no information to the current analysis.

Phone numbers and identifiers are stored as text rather than numeric values. Telephone fields can contain values such as `anonymous`, and numeric conversion can also remove leading zeros or mishandle international prefixes. Duration is assigned a numeric type, while dates and timestamps receive the appropriate temporal types. Account text is cleaned, trimmed and converted to lowercase before matching. These preparation steps reduce avoidable matching errors and provide consistent fields for the relationships and calculations described in the next section.

3 Analytical approach and Power BI implementation

3.1 Separating data preparation from calculations

The implementation separates data preparation, model relationships and reporting measures. Power Query imports the workbooks, cleans fields, selects the relevant records and creates classification columns. Relationships connect the prepared tables so that date, team and staff selections filter the appropriate records. Data Analysis Expressions (DAX) measures then calculate the values shown in the report. This separation makes it easier to investigate an unexpected result: the issue may come from the source data, a preparation step, a relationship or the measure itself.

3.2 Identifying Main IVR entries and related records

The inbound model starts with `FactDirectRouting`, which contains the prepared inbound records. Power Query renames the source UPN field to `ResourceAccount`. A record is classified as a Main IVR entry when the cleaned account equals `aa-mainivr-v2@ahri.com.au` and its CorrelationId equals its Shared correlation ID. The account condition identifies the main-menu resource, while the identifier condition selects the entry record used by the measure. In this export, the rule produces 3,986 entries.

A second table, `FactRoutingGroup`, groups the inbound records by Shared correlation ID. Each group receives indicators showing whether it contains Main IVR, queue, overflow-named, user-associated or unsuccessful records. Its entry date is the earliest observed Invite time within the group, converted to Melbourne time. The resulting table has one row per related-record group. This supports group-level summaries while retaining the individual records in `FactDirectRouting` for more detailed analysis.

The two tables answer different questions. Counting source rows describes recorded system activity. Counting shared-ID groups combines related records. Restricting the groups to those containing Main IVR then applies the agreed inbound scope. The reduction from 9,859 records to 6,361 groups is therefore a grouping operation, while the reduction to 3,986 Main IVR groups is a scope decision. Neither step should be described as removing thousands of duplicate or invalid calls.

The Main IVR entry count equals the number of groups containing the Main IVR account in this export. This agreement provides a useful check, but it should be tested again with future data. The model does not assume that every possible export will contain exactly one qualifying entry per group. Keeping separate entry and group measures makes their counting rules visible and allows differences to be investigated.

3.3 Classifying queue destinations

Queue classification uses the presence of "Queue" in Display Name. A queue record is additionally classified as overflow when its name contains "Overflow". Service names are derived from these labels so that primary and overflow destinations can be compared within the same service. This approach uses information available in the export, but depends on consistent account naming. A maintained account-to-service mapping would be more reliable if names or routing arrangements change.

Main IVR groups are assigned to three mutually exclusive categories. A group containing an overflow-named queue record is classified as "Overflow queue observed". A group containing queue records but no overflow-named record is classified as "Primary queue only observed". The remaining groups are classified as "No queue record observed". These categories describe the available records. The absence of a queue record does not establish that the caller abandoned the menu or that the call ended at that point.

The queue measure counts distinct shared IDs associated with queue records in the selected service and period. Overflow destination share divides groups containing overflow-named records by groups containing any queue record. Primary and overflow counts can overlap when the same group contains both types of record. The percentage therefore describes recorded destination associations, rather than a confirmed probability of escalation.

The percentage chart displays a service only when it has at least thirty queue-associated groups in the current selection. Here, `n` means the number of observed groups, and the rule is `n >= 30`. Thirty is a display threshold chosen during development, not a value extracted from Excel or a telephone queue capacity. It reduces attention to percentages based on

very small counts, but does not guarantee statistical reliability. Its sensitivity is considered in the evaluation section.

3.4 Selecting and counting outbound activity

Outbound preparation starts from `RawCallLogs`, filters Call Direction to Outbound and standardises the UPN before matching it against `DimStaff`. An inner join retains records associated with the approved staff accounts. The nested match column is then removed because staff names and teams remain available through the model relationship. The mapping contains one row per UPN, supporting an unambiguous account-to-team assignment and the one-to-many relationship required by the model.

The resulting `FactOutbound` table contains 6,046 records and 6,046 distinct correlation IDs. The main outbound measure uses a distinct count of CorrelationId to make its counting unit explicit. This equals the row count in the current selected dataset. A separate measure counts distinct staff accounts with recorded outbound activity. That measure describes observed participation within the selection, rather than the number of employees available or rostered to work.

For example, selecting one week may reduce the staff count because some approved staff have no recorded attempts during that week. Those staff remain in the approved mapping, but do not appear in the activity-based count. This distinction prevents the dashboard from presenting a changing activity count as a changing team headcount. Team and staff selections are applied through `DimStaff`, allowing the same measures to be reused consistently across different views.

3.5 Time conversion and average activity

Invite timestamps are parsed as UTC and converted to Melbourne time before the date and hour fields are created. The Power Query logic uses date-based daylight saving rules to select an offset of ten or eleven hours. The supplied reporting period falls within the ten-hour offset. Extracting the date after conversion is important because an evening UTC timestamp can fall on the following calendar day in Melbourne. Using the original UTC date would therefore misclassify some daily and hourly activity.

The date table contains every date between the observed inbound boundaries. Average weekday activity is calculated over selected Monday-to-Friday dates that pass the complete-day rule. Dates with zero activity remain in the denominator, while the first and last observed dates are excluded. Public holidays remain included because the model does not

contain a verified AHRI operating calendar. The measure should therefore be understood as an average per calendar weekday, rather than per confirmed working day.

Across the full export, there are 105 eligible weekdays. These contain 3,925 Main IVR entries and 6,027 selected outbound attempts. Dividing each count by 105 gives 37.4 inbound entries and 57.4 outbound attempts per weekday. These numerators differ from the full-period totals because the averages use a narrower set of dates. The monthly weekday comparison applies the same principle, allowing months with different numbers of weekdays to be compared more fairly.

The model also defines a trailing seven-day average. For each date, it divides the Main IVR entries from that date and the preceding six calendar days by seven. Each window includes one occurrence of every weekday and retains zero-activity days. This reduces short-term variation while preserving the broader movement in activity. Only windows entirely within the accepted source boundaries are included. The calculation can retrieve preceding dates outside a narrow date selection, so selecting one date does not automatically reduce its window to one day.

3.6 Technical outcomes and duration

Technical outcome measures use the source's Yes and No values. The outbound technical success rate divides successful attempts by all selected attempts. Mean and median duration are calculated from technically successful outbound records, including recorded zero durations. The mean describes the arithmetic average, while the median identifies the middle observation and is less affected by unusually long durations. These measures describe recorded technical activity, they are not treated as staff talk time or evidence that a customer's enquiry was resolved.

3.7 Relationships and reproducibility

The date table, `DimRoutingDate`, has active relationships with the date fields in `FactDirectRouting`, `FactRoutingGroup` and `FactOutbound`. `DimStaff` connects to `FactOutbound` through UPN. Filtering operates from the dimension tables towards the fact tables. This follows the dimensional modelling principle of separating attributes used for filtering from observations used for summarisation (Microsoft, 2024). During development, a missing outbound date relationship prevented daily filters from reaching the outbound records. Creating that relationship corrected the weekday-average behaviour.

Both workbook queries use the `ProjectRoot` parameter to construct their source paths. Collaborators can therefore update their local project location without rewriting each query.

The Power BI project format stores report and model definitions in text files, supporting version control and inspection of changes (Microsoft, 2026b). Separate checks against the Excel workbooks provide an additional comparison for the main counts. These checks support the intended calculation rules, but do not replace testing the rendered report and its interactive filters in Power BI Desktop.

3.8 Dashboard navigation and use

The five implemented Power BI pages are reproduced in Appendix A. Date slicers allow users to examine a selected reporting period. The outbound pages also provide team and staff filters, and the activity page supports drilling from team totals to staff accounts. These controls allow AHRI to investigate patterns within the agreed reporting scope.

The inbound and outbound outcome pages (Figure 10 and Figure 12) provide additional technical context. Their status labels and duration measures require the interpretation described in this report: they do not establish customer resolution, waiting time or individual productivity.

4 Inbound findings

4.1 Overall demand and the comparison period

AHRI's main telephone menu received 3,986 recorded Main IVR entries between 6 April and 1 September 2026. These entries describe activity through the main menu rather than every call received by AHRI. Repeated calls from the same person can contribute multiple entries.

The timing analysis excludes the first and last observed dates because their coverage may be incomplete. This leaves 147 calendar days, including 105 Monday-to-Friday dates. Those weekdays contain 3,925 entries, giving an average of 37.4 per weekday. The median is 33, meaning that a typical weekday has fewer entries than the arithmetic average suggests. Higher-volume days pull the average upward.

Dates with no entries remain included. Public holidays also remain classified by their calendar weekday. The analysis assumes that the export provides continuous coverage between its boundary dates. The records alone cannot prove that every day was captured completely.

The corresponding Power BI overview is shown in Figure 8 in Appendix A. It brings the entry totals, daily trend, trailing seven-day average and timing comparisons together in one interactive page.

4.2 Daily variation and unusually busy periods

Figure 1 shows that demand varies considerably between days. The two largest daily counts occur consecutively: 130 entries on 20 May and 134 on 21 May. Together, these dates account for 23.7% of May's weekday entries. This concentration suggests that the month's high average deserves investigation alongside AHRI's communication, renewal and event calendars.

The trailing seven-day average helps distinguish a single busy day from a broader period of activity. On 21 May, it is 55.3 entries per calendar day, calculated from 387 entries across 15–21 May. This includes the weekend within that window. It therefore answers a different question from the average per weekday, which excludes Saturdays and Sundays.

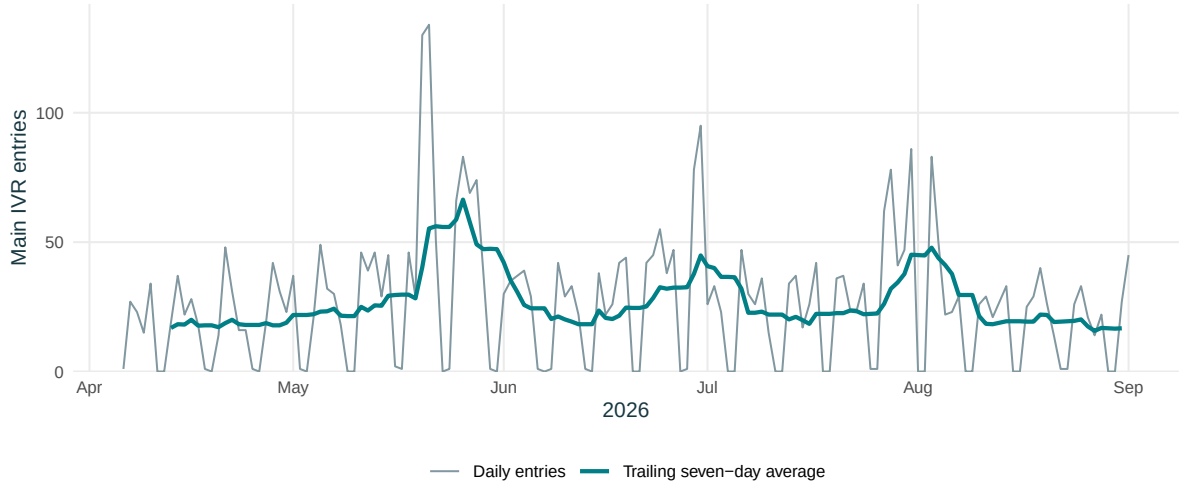


Figure 1: *Daily Main IVR entries and trailing seven-day average in Melbourne time. The moving average includes weekends and zero-entry days. Only windows excluding the first and last observed dates are shown.*

These peaks are observations to explain, rather than errors to remove automatically. As a sensitivity check, excluding 20 and 21 May reduces May's weekday average from 53.0 to 44.6. May remains busier than August even under this alternative calculation. The comparison shows that the two peaks influence the size of the decline but do not fully explain it.

4.3 Changes across fully covered months

Average weekday demand decreases from 53.0 in May to 39.5 in June, 37.4 in July and 29.5 in August. August's average is approximately 44.2% below May's. The monthly medians also decline, from 46 to 38, 34 and 26 respectively, supporting a broader reduction beyond the effect of a few exceptionally busy days.

Table 2 accounts for the different numbers of weekdays in each month. It also shows that the busiest named weekday changes across months.

Table 2: *Main IVR demand during fully covered months. Means and medians refer to daily counts on Monday-Friday dates. The busiest day is the named weekday with the highest average within that month.*

Month	Weekdays	Mean	Median	Busiest day
May	21	53.0	46	Wednesday
Jun	22	39.5	38	Tuesday
Jul	23	37.4	34	Tuesday
Aug	21	29.5	26	Monday

This remains a descriptive comparison over four full months. It does not establish a recurring seasonal pattern or predict future demand. A reduction in calls could reflect quieter activity, different contact channels or changes in the reasons members call. Information about campaigns and a longer history would be needed to distinguish these explanations.

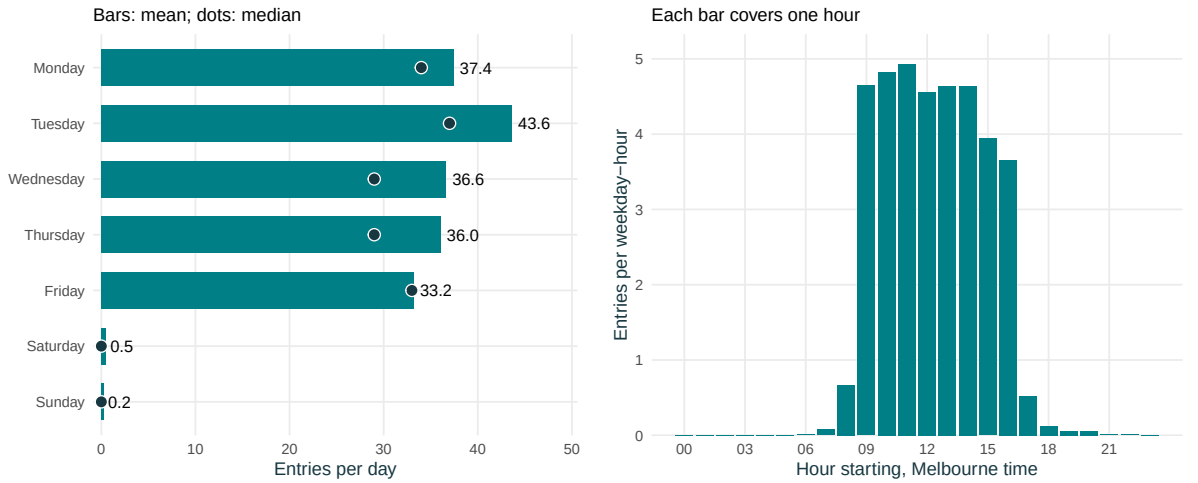
4.4 Differences between days of the week

Tuesday has the highest overall average, with 43.6 entries per eligible Tuesday. Monday averages 37.4, Wednesday 36.6, Thursday 36.0 and Friday 33.2. Tuesday's average is approximately 31% higher than Friday's. Each named weekday occurs 21 times in the eligible period, so these comparisons use equal numbers of dates.

The medians add useful context, as shown in panel (a) of Figure 2. Tuesday also has the highest median, at 37 entries, indicating that its higher average is not solely associated with one unusually busy date. However, Friday's median of 33 exceeds Wednesday's and Thursday's medians of 29. The day with the lowest average is therefore not necessarily the day with the lowest typical count.

Only 15 entries occur on eligible weekends: ten on Saturdays and five on Sundays. Both weekend medians are zero. This indicates that observed demand is strongly concentrated on weekdays, although the export does not explain AHRI's opening arrangements or the needs of those weekend callers.

Tuesday should not be treated as the busiest day in every month. Wednesday has the highest weekday average in May, while Monday leads in August. The overall Tuesday pattern provides a useful starting point for reviewing coverage, but the changing monthly pattern argues against basing a permanent roster on the overall ranking alone.



(a) *Mean and median daily entries.*

(b) *Mean entries within each hour.*

Figure 2: *Weekday and hourly Main IVR patterns in Melbourne time. Panel (a) uses 21 occurrences of each named day. Panel (b) divides each hourly total by all 105 eligible Monday-Friday dates, including dates with no entries in that hour.*

4.5 When during the day calls arrive

Panel (b) of Figure 2 shows a broad daytime concentration. Of the 3,925 eligible weekday entries, 3,764 occur between 09:00 and 17:00, representing 95.9%. There are 79 entries before 09:00 and 82 from 17:00 onward.

The highest hourly average occurs from 11:00 to 12:00, at 4.9 entries per eligible weekday. However, this is only slightly above the surrounding hours. From 09:00 to 15:00, hourly averages remain between approximately 4.6 and 4.9 entries. The main pattern is therefore a sustained period of daytime activity, rather than a narrow peak requiring attention at only one hour.

There is no pronounced lunchtime reduction. The averages for 12:00–13:00 and 13:00–14:00 are 4.6 entries each. This supports reviewing whether breaks and other duties leave sufficient telephone coverage around lunchtime. Average demand then falls to 3.9 entries during 15:00–16:00 and 3.7 during 16:00–17:00, although these hours still contain regular activity.

These timestamps record entry into the telephone system. They do not measure when an employee answered, how long callers waited or how many calls were being handled simultaneously.

4.6 Combining weekday and hour

The separate weekday and hourly profiles can conceal differences between particular parts of the week. Figure 3 combines them by calculating average entries for every weekday-hour combination. For example, Tuesday's 10:00–11:00 value divides all entries in that slot by the 21 eligible Tuesdays.

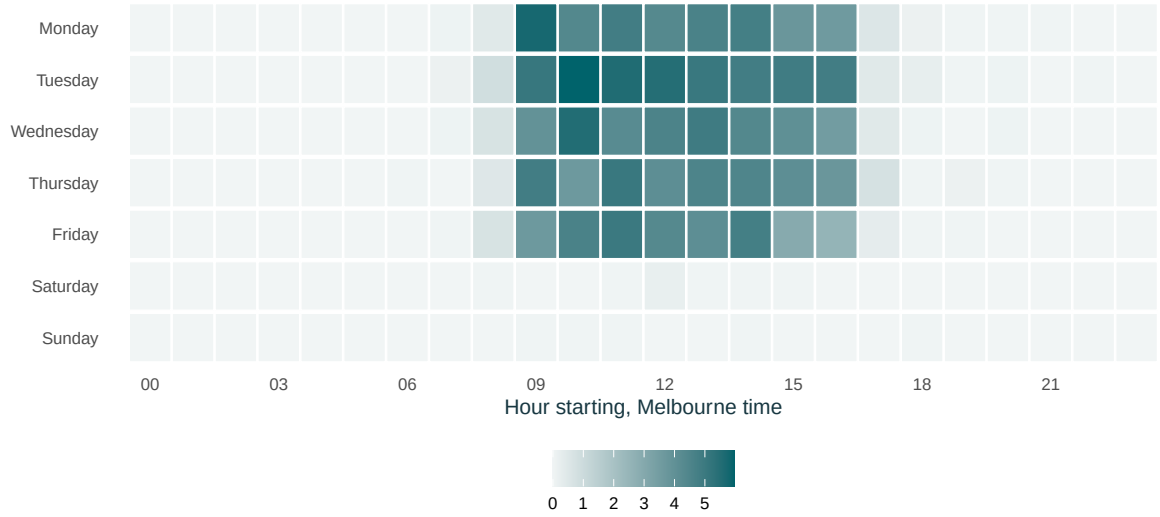


Figure 3: *Average Main IVR entries by weekday and hour in Melbourne time. Each cell uses all 21 occurrences of that named day, including dates with no entries in the relevant hour. Darker cells indicate higher historical average demand.*

The highest average cell is Tuesday from 10:00 to 11:00, at approximately 6.0 entries. Monday from 09:00 to 10:00 also stands out, averaging 5.7. Tuesday's relatively high activity extends across several morning and midday hours, which helps explain its larger daily average.

This view supports more specific coverage discussions than a daily total alone. AHRI could compare meeting schedules, breaks and staff availability with these periods. Nevertheless, an average of six entries in an hour does not imply six simultaneous calls or a requirement for six staff. Staffing decisions also require handling times, concurrent demand, available staff and service targets.

4.7 Recorded queue destinations

Across the full observed period, 1,262 of the 3,986 Main IVR groups contain a queue record, while 2,724 do not. Every queue-associated group in this export also contains Main IVR. These classifications describe recorded activity; the absence of a queue record does not establish that a caller abandoned the call or failed to receive assistance.

Among queue-associated groups, 337 contain an overflow-named destination, equivalent to 26.7%. The other 925 contain primary queue records without an observed overflow destination.

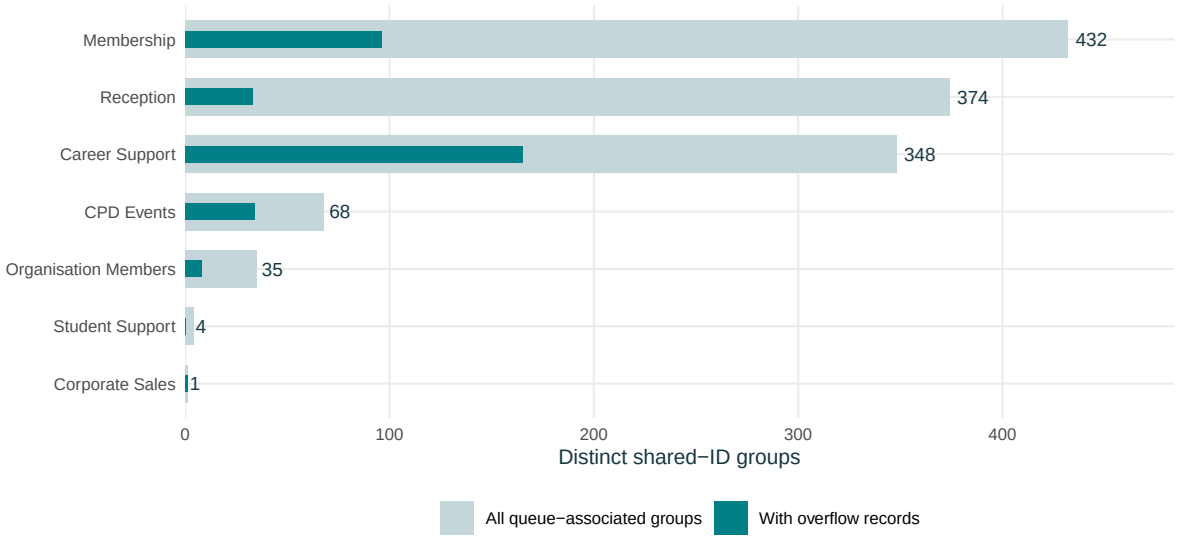


Figure 4: *Distinct queue-associated groups by service across the full observed period. Darker bars show the overflow-associated subset of each service total. The bars are not additive.*

Membership receives the largest number of queue-associated groups, with 432, followed by Reception with 374 and Career Support with 348. CPD Events has 68, Organisation Members 35, Student Support four and Corporate Sales one. The counts in Figure 4 provide the context needed to interpret overflow percentages.

Among services meeting the dashboard's minimum of 30 groups, CPD Events has the highest overflow share, with 34 of 68 groups, or 50.0%. Career Support follows with 165 of 348, or 47.4%. Organisation Members records 22.9%, Membership 22.2% and Reception 8.8%. The minimum of 30 is a display rule for queue-percentage comparisons, not a telephone-system setting or a guarantee of statistical reliability.

These differences justify reviewing the configuration of the named destinations. They do not establish that one service performs worse than another. In particular, CPD Events' 50% share does not prove that half its callers waited too long. The reasons for using each overflow account need confirmation from AHRI's telephone administrator.

The implemented routing page is reproduced in Figure 9. It presents service counts alongside overflow destination shares, with the minimum of 30 observed groups applied to the percentage comparison.

4.8 Implications for coverage and service review

The findings support three practical priorities. First, review weekday coverage across the sustained 09:00–15:00 period, including lunchtime, while checking whether Tuesday mornings and Monday opening hours coincide with reduced staff availability. Second, investigate the exceptionally busy dates in May and compare the changing monthly pattern with planned communications and events. Third, confirm the routing arrangements behind the relatively high overflow shares for CPD Events and Career Support.

These are evidence-based starting points for investigation. The analysis identifies when demand is concentrated and where recorded routing deserves attention. Determining whether coverage is sufficient requires information about waiting, handling time and available staff that the current export does not provide.

5 Outbound findings

5.1 Connecting incoming demand with staff calling activity

The inbound analysis established when calls arrive through AHRI's main menu and which recorded destinations receive that demand. Outbound analysis adds a complementary view: when the approved staff make calls, how that activity is distributed across teams and what technical outcomes are recorded. Together, these views provide a broader basis for discussing telephone coverage and the scheduling of outgoing calls.

For example, periods with substantial incoming and outgoing activity may warrant a review of staff availability and competing responsibilities. Understanding both directions helps managers identify those periods rather than assuming that incoming calls represent all telephone activity.

The two analyses cover different populations. Inbound findings concern Main IVR activity, while outbound findings concern the 18 staff accounts approved by AHRI. Their counts cannot be combined into a count of customers or treated as a complete measure of staff workload. The export also does not establish whether an outbound attempt follows an earlier inbound enquiry, belongs to a campaign or serves another purpose.

The comparison therefore focuses on the timing and distribution of recorded activity. It identifies questions for operational review while leaving explanations about staffing, campaigns and customer journeys to additional information from AHRI.

5.2 Activity within the approved staff group

The 18 approved staff recorded 6,046 outbound attempts between 6 April and 1 September 2026. Membership accounts for 5,275 attempts, or 87.2% of the selected total. HRSC accounts for 680, or 11.2%, and Events accounts for 91, or 1.5%. The percentages total slightly less than 100% because of rounding.

The raw export contains 7,796 outbound records. Of these, 1,750 fall outside the approved account list and were excluded to maintain the agreed scope. The results therefore describe the selected staff group rather than AHRI's entire outbound activity. Table 3 summarises the selected attempts and source-recorded success status. All 18 approved accounts appear in the observed records.

Table 3: *Outbound attempts and recorded technical success by team, 6 April to 1 September 2026. Success refers to the source status and does not establish a completed human conversation.*

Team	Staff	Attempts	Success = Yes	Success (%)
Membership	11	5,275	4,765	90.3
HRSC	5	680	654	96.2
Events	2	91	85	93.4
Total	18	6,046	5,504	91.0

Membership has both the largest approved staff group and the largest number of attempts. However, headcount alone cannot explain the difference between teams. The export does not contain working hours, leave, campaign assignments or the proportion of each person's role devoted to telephone work. Dividing attempts by staff numbers would produce an arithmetic average, but it would not provide a fair comparison of employee productivity.

The team breakdown is most useful for locating recorded activity and identifying questions for managers. AHRI could investigate whether Membership's larger volume reflects scheduled renewal activity or a greater responsibility for outbound contact. These explanations require operational information beyond the call logs. A smaller team total should not automatically be interpreted as poor performance.

Figure 11 shows how these totals are presented in Power BI alongside date, team and staff filters, daily activity, monthly averages and a weekday-by-hour view.

5.3 Timing and monthly changes

The selected staff average 57.4 outbound attempts per eligible weekday. This uses 6,027 attempts across 105 weekdays, with the remaining 19 attempts occurring on the excluded final boundary date. No outbound attempts are recorded on weekends for the selected accounts.

Two eligible weekdays contain no attempts. Keeping these dates in the calculation measures activity across the complete eligible calendar. Calculating the average only on days with at least one attempt would increase it to 58.5. This demonstrates why the denominator matters: an average over active days answers a different question from an average over all eligible weekdays.

Outbound activity varies considerably across fully covered months. May records 1,634 attempts, averaging 77.8 per weekday. June records 857, averaging 39.0. July records 1,325, averaging 57.6, and August records 1,089, averaging 51.9. The July increase means the pattern is not a continuous month-by-month decline.

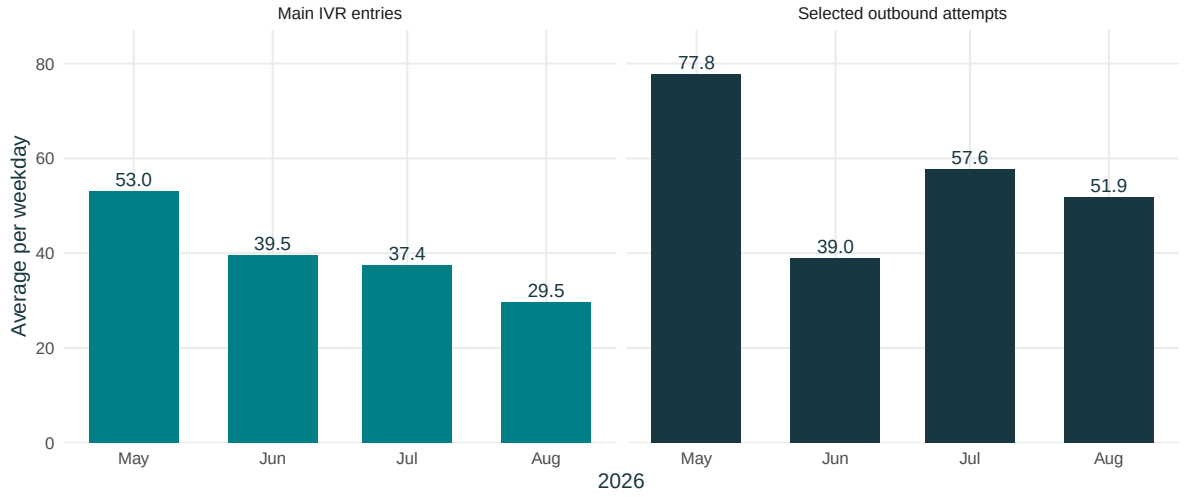


Figure 5: *Average activity per weekday for fully covered months. Both panels use the same weekday denominator and vertical scale. Inbound entries and outbound attempts describe different populations and should not be added into a count of customers.*

Figure 5 compares the timing of incoming demand and outgoing activity using the same weekday denominator. This makes their patterns easier to compare, although the different populations mean that bar heights should not be interpreted as equivalent amounts of work.

Both measures are highest in May among the four fully covered months. Their subsequent movements differ: inbound weekday averages decrease from May through August, while outbound activity falls in June, recovers in July and declines again in August. The July

recovery shows why the two directions need separate analysis. A reduction in incoming calls does not necessarily coincide with a reduction in calls made by staff.

For AHRI, these differences provide a starting point for reviewing how telephone activity is organised. Managers could compare the observed periods with campaign schedules, staff availability and other responsibilities. Where outgoing activity is planned, this information may help identify periods worth considering for that work. However, quieter inbound periods do not establish spare capacity because available staff, handling time and other duties are not measured here.

The comparison connects the two views without assuming that one causes the other. Confirming whether staff shifted effort between incoming and outgoing calls would require additional operational evidence.

5.4 Technical outcomes and recorded duration

The export marks 5,504 selected attempts as successful and 542 as unsuccessful, giving a technical success rate of 91.0%. Microsoft's documentation distinguishes fully established calls from failed calls (Microsoft, 2025). The export does not establish whether the intended person answered or whether the conversation achieved its purpose. The report therefore treats success as technical status rather than a customer answer rate.

All 542 unsuccessful attempts have zero recorded duration and missing start timestamps. This consistent pattern supports retaining them as attempts while calculating successful-call duration separately. Including unsuccessful attempts in that duration average would change the population being described and reduce the result.

Among technically successful attempts, mean recorded duration is 95.8 seconds and median duration is 41 seconds. The 90th percentile is 218 seconds, meaning approximately 90% of these durations are at or below that value. The mean being substantially higher than the median indicates that longer observations pull the average upward.

The most frequent duration category is 31–60 seconds, as shown in Figure 6. Approximately 69.4% of technically successful attempts last no more than 60 seconds. There are also 26 successful records with zero duration. These remain included because their short duration alone does not establish that they are invalid. Their meaning should be checked with AHRI's IT team.

Recorded duration describes one aspect of telephone activity. It does not capture preparation, follow-up work or time spent updating other systems. Consequently, neither a higher

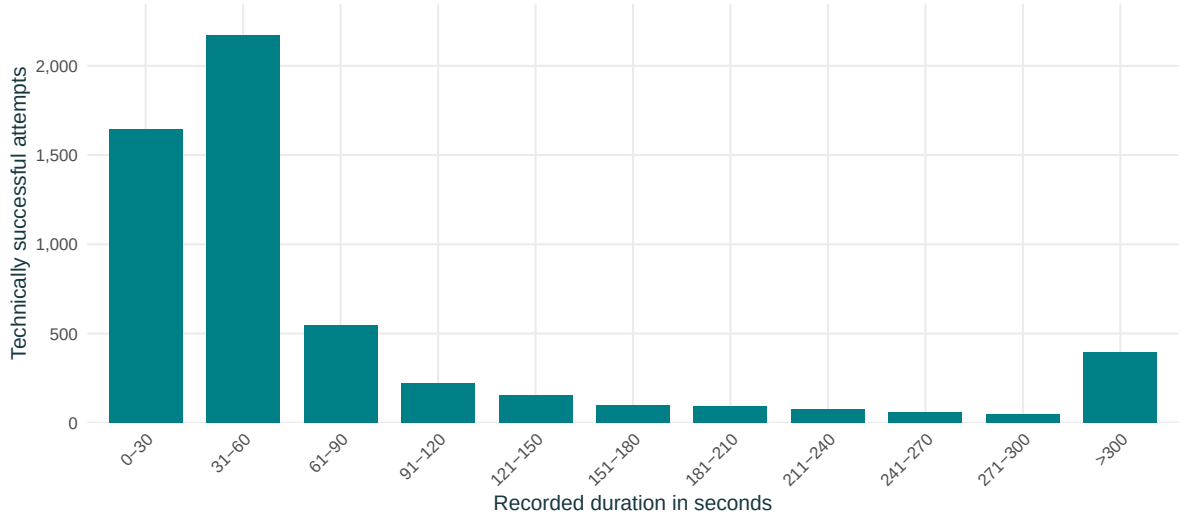


Figure 6: *Recorded duration of the 5,504 technically successful outbound attempts. Bars show counts within duration categories. The final category includes every duration above 300 seconds and is not an equal-width numerical interval.*

call count nor a longer average duration is sufficient evidence of greater productivity.

The dashboard's readable outcome labels also require care. It currently places SIP code 487 in the category "Not answered", but the protocol defines this code as "Request Terminated" (Rosenberg et al., 2002). There are 365 unsuccessful attempts with this code in the selected data, so its interpretation affects a substantial part of the outcome analysis.

5.5 What the combined analysis provides

Taken together, the inbound and outbound findings give AHRI two complementary views of its telephone operations. The inbound results identify the timing of demand arriving through the main menu. The outbound results show when the approved staff initiate calls, where those attempts are concentrated and how their recorded outcomes vary.

Their combined value is to support more focused management questions: does telephone coverage match incoming demand, how does planned outgoing activity fit alongside that demand, and which technical outcomes require investigation? Answering these questions fully requires staff availability, campaign information and confirmed outcome definitions.

The dashboard provides the evidence needed to begin those discussions. Its contribution is a clearer account of recorded activity and the questions that activity raises, rather than an assumption that call counts alone explain operational performance.

6 Statistical summaries and variability

6.1 Why averages need context

An average describes the level of activity but can hide differences between quiet and busy days. This section adds standard deviation (SD), medians, quartiles and boxplots. SD describes spread around the mean in the original units. Quartiles are less influenced by extreme observations and provide a useful comparison when distributions have long tails (Diez et al., 2019).

Daily summaries use the 105 eligible Monday-to-Friday dates from 7 April to 31 August 2026. The first and last observed dates are excluded because coverage may be partial. Public holidays and zero-activity weekdays remain included, assuming the export is complete between these boundaries. Duration summaries use all 5,504 technically successful outbound attempts in the selected staff population.

The sample SD is calculated as:

$$s = \sqrt{\frac{\sum_{i=1}^{n}(x_i - \bar{x})^2}{n-1}},$$

where x_i is a daily count or recorded duration, $\bar{x}$ is its mean, and n is the number of observations. This follows R's `sd()` convention (R Core Team, n.d.), it does not imply that the export is a random sample.

6.2 Overall activity and duration

Table 4: *Observed variation in daily activity and successful outbound duration. For activity, n counts weekdays; for duration, n counts attempts. Q1 and Q3 are the 25th and 75th percentiles.*

Metric	n	Mean (SD)	Median [Q1, Q3]
Daily Main IVR entries	105	37.4 (21.7)	33.0 [24.0, 42.0]
Daily outbound attempts	105	57.4 (38.8)	48.0 [26.0, 80.0]
Successful duration (s)	5,504	95.8 (184.3)	41.0 [28.0, 73.0]

Main IVR activity averages 37.4 entries per weekday, with an SD of 21.7 (Table 4). The median is 33, and the middle half of daily counts lies between 24 and 42. Outbound activity averages 57.4 attempts, with an SD of 38.8 and a median of 48. Both measures show that the average alone is an incomplete guide to daily activity.

Successful outbound duration averages 95.8 seconds, with an SD of 184.3 seconds. Its median is 41 seconds, and its quartiles are 28 and 73 seconds. The long upper tail pulls the mean upwards. Reporting both summaries preserves the contribution of longer records while showing what is typical. Recorded duration does not establish staff handling time.

6.3 How activity varies within each weekday

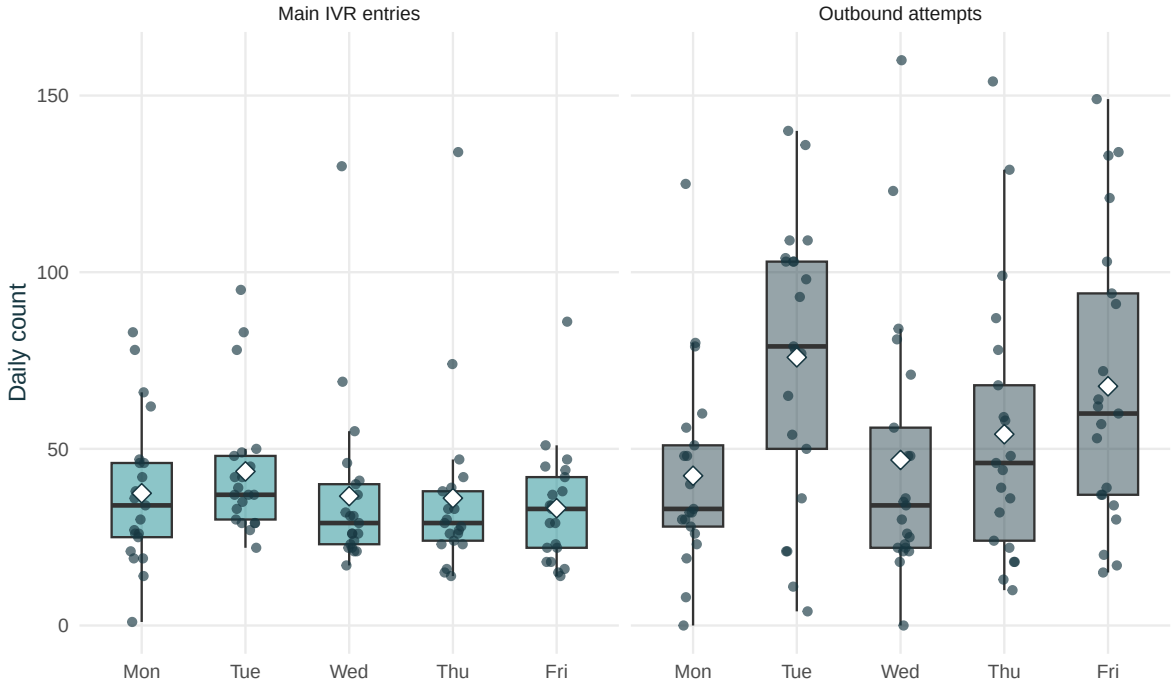


Figure 7: *Daily activity by weekday, with 21 dates in each boxplot. Dots represent individual dates, horizontal lines show medians, and white diamonds show means. Both panels use the same count scale.*

Each box in Figure 7 covers the middle 50% of observations. Whiskers extend to the most extreme observations within 1.5 interquartile ranges of the box; observations beyond them are retained (Diez et al., 2019). They identify unusual days for investigation, rather than proving that records are incorrect. All dates are plotted, with a small horizontal offset to reduce overlap. The boxplots were produced in R using ggplot2 (Wickham, 2016)..

Tuesday has the highest inbound mean, at 43.6 entries, with an SD of 19.2. Wednesday and Thursday have lower means but larger SDs of 24.9 and 26.0, influenced by the May peaks. Lower average demand therefore does not guarantee consistency. The outbound panel also shows substantial variation within weekdays, so an average should not be treated as a fixed daily target.

6.4 Variation across fully covered months

Table 5: *Daily activity within fully covered months, shown as mean (SD). Each SD is calculated across individual weekdays in that month.*

Month	Weekdays	Main IVR: mean (SD)	Outbound: mean (SD)
May	21	53.0 (31.3)	77.8 (33.5)
Jun	22	39.5 (19.0)	39.0 (31.1)
Jul	23	37.4 (17.6)	57.6 (42.8)
Aug	21	29.5 (14.6)	51.9 (31.9)

May's inbound mean and SD are 53.0 and 31.3, by August they are 29.5 and 14.6 (Table 5). The whole-period average therefore combines periods with different activity levels and variability. Outbound activity differs: July has the largest monthly SD, although May has the highest mean.

6.5 What these summaries can establish

SD and boxplots describe observed variation. A confidence interval instead describes uncertainty in an estimated mean (Diez et al., 2019). Neither SD nor a confidence interval for the mean is a forecast range. This report focuses on the observed export period. A conventional confidence interval assuming independent observations is not added because daily counts may be dependent and activity changes over time. Forecasting would require a time-series model and prediction intervals. These descriptive comparisons support reviewing coverage and calling plans, but do not establish statistically significant differences or optimal staffing levels.

7 Conclusion

This project developed a Power BI dashboard to help AHRI understand its telephone activity using the call records available. The main analytical challenge was establishing what those records represented. Grouping related records and agreeing on the reporting scope made it possible to distinguish Main IVR demand from calls made directly to staff. The resulting analysis covers 3,986 Main IVR entries and 6,046 outbound attempts by the 18 approved staff accounts between 6 April and 1 September 2026.

The inbound findings identify periods worth considering when reviewing telephone coverage. Activity is concentrated on weekdays, with sustained demand throughout the morning and early afternoon, including lunchtime. Tuesday has the highest overall weekday average,

although the busiest weekday changes between months. Average weekday demand also decreases between May and August. These patterns provide a starting point for reviewing rosters, breaks and meetings, while the unusually busy dates in May warrant comparison with AHRI's communications and event calendar.

Outbound activity adds a complementary view of calls initiated by staff. Most selected attempts belong to Membership, and monthly activity follows a different pattern from inbound demand. Examining both directions helps AHRI consider incoming demand alongside planned calling activity. However, the records cannot establish whether an outbound attempt responds to a particular inbound enquiry or how much staff capacity remains available.

The additional statistical summaries strengthen the interpretation of these findings. Standard deviations and boxplots show substantial variation between dates, while medians and quartiles provide context where unusually large observations influence the mean. Average activity should therefore be considered alongside its distribution and changes over time when discussing operational plans.

The dashboard supports these discussions within the limits of the available data. Technical success does not confirm a conversation or a resolved enquiry, and recorded destinations do not establish waiting times or abandonment. Further work should prioritise confirming routing and outcome definitions, maintaining consistent checks during refreshes, and comparing the findings with staff availability and campaign information. The project's contribution is a reproducible view of recorded telephone activity that gives AHRI a clearer basis for investigation and planning.

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Appendix A: Power BI dashboard

Source: Authors' Power BI dashboard. Selected dates: 6 April to 1 September 2026.

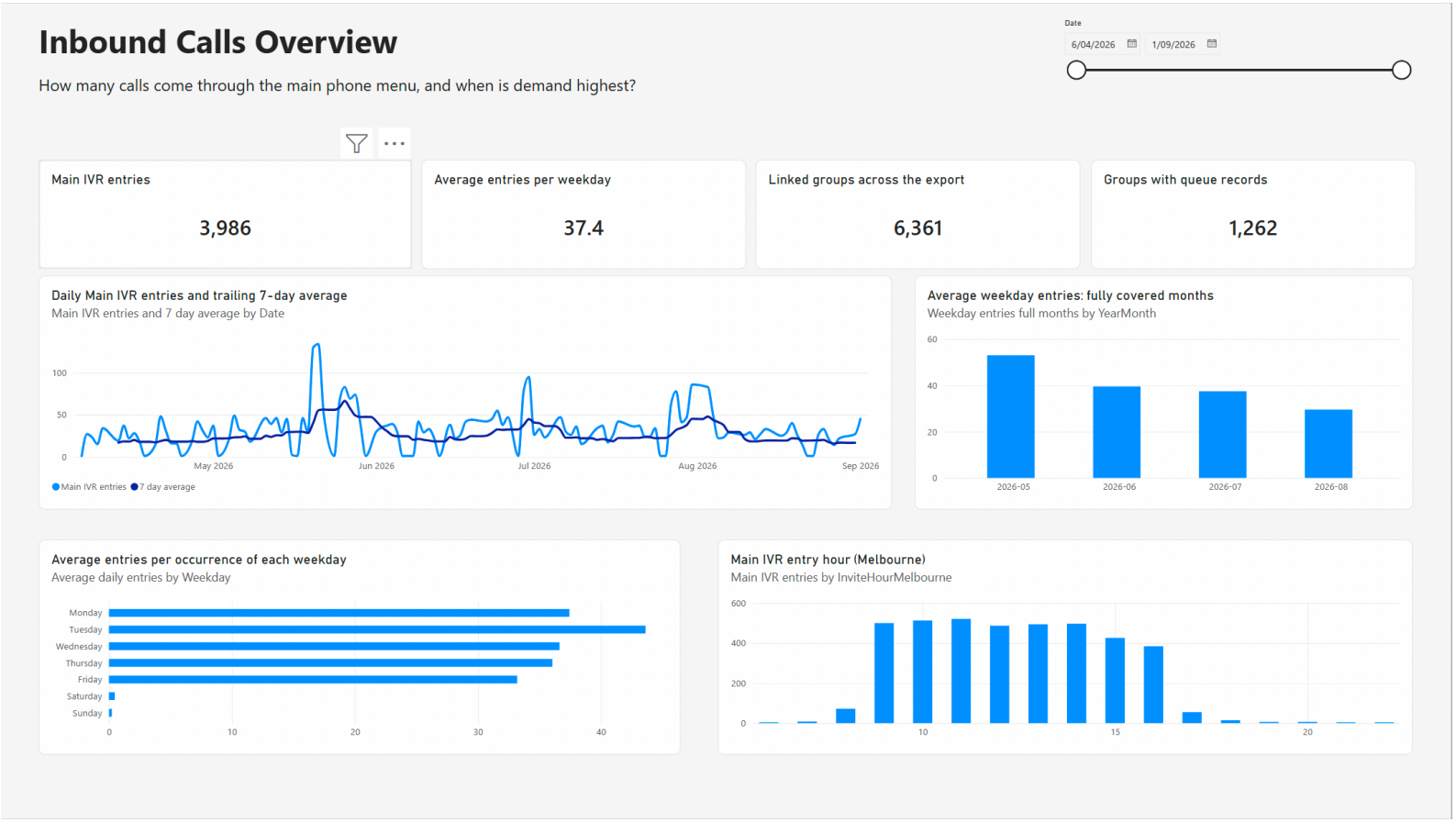


Figure 8: *Inbound Calls Overview. The date slicer connects Main IVR entry totals with daily, weekday, hourly and monthly views. The trailing seven-day line smooths daily variation. Linked groups include records outside the Main IVR scope.*

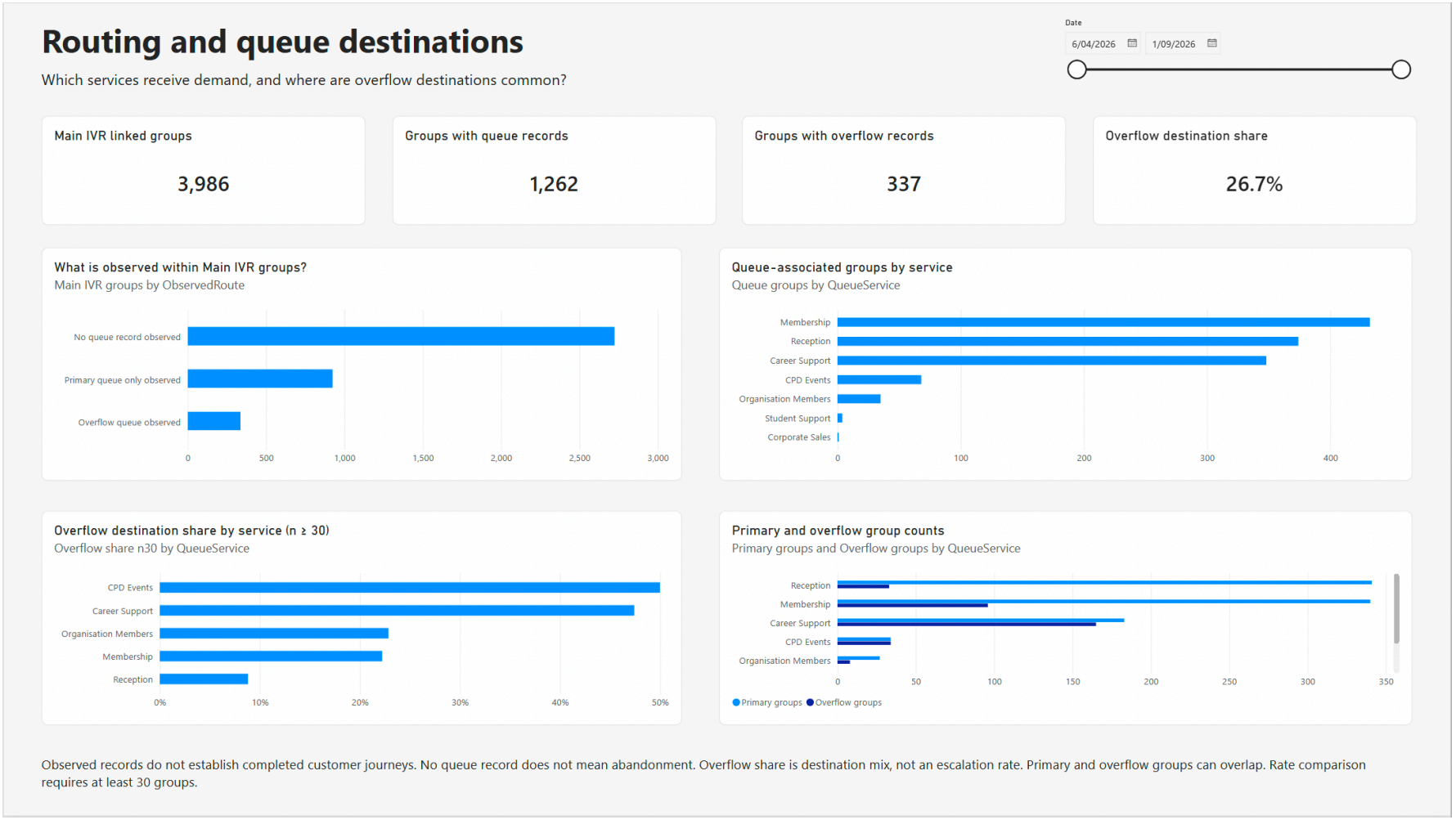


Figure 9: *Routing and queue destinations. The page compares service volumes and observed destinations. The $n \geq 30$ rule restricts percentage displays; it is not queue capacity. Overflow labels and missing queue records do not establish escalation or abandonment.*

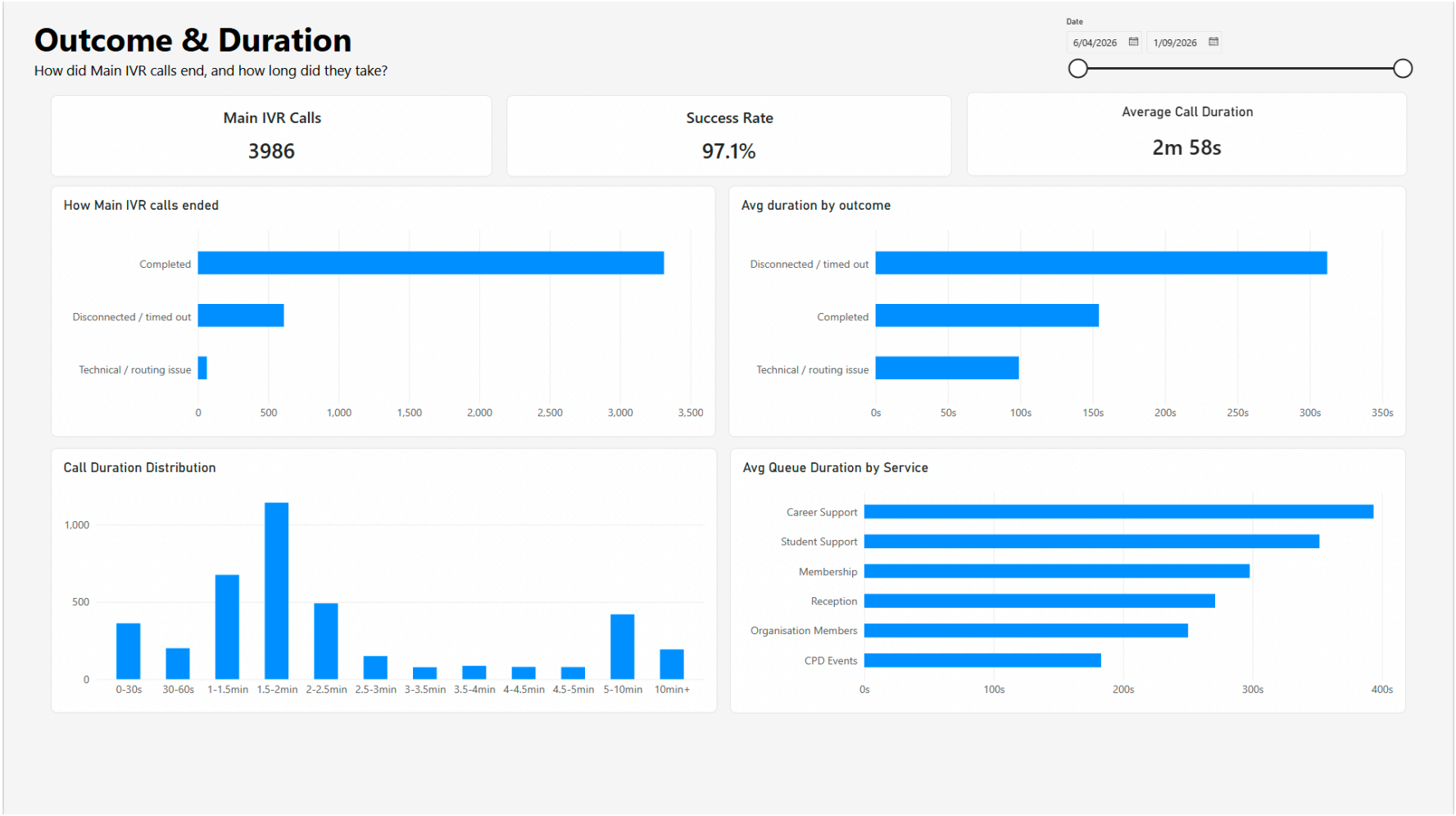


Figure 10: *Inbound outcome and duration. The success card uses final-record technical status; the outcome bars use a separate classification of system codes and phrases. The duration card averages summed record durations within linked groups. These measures do not establish enquiry resolution, caller waiting time or agent talk time.*

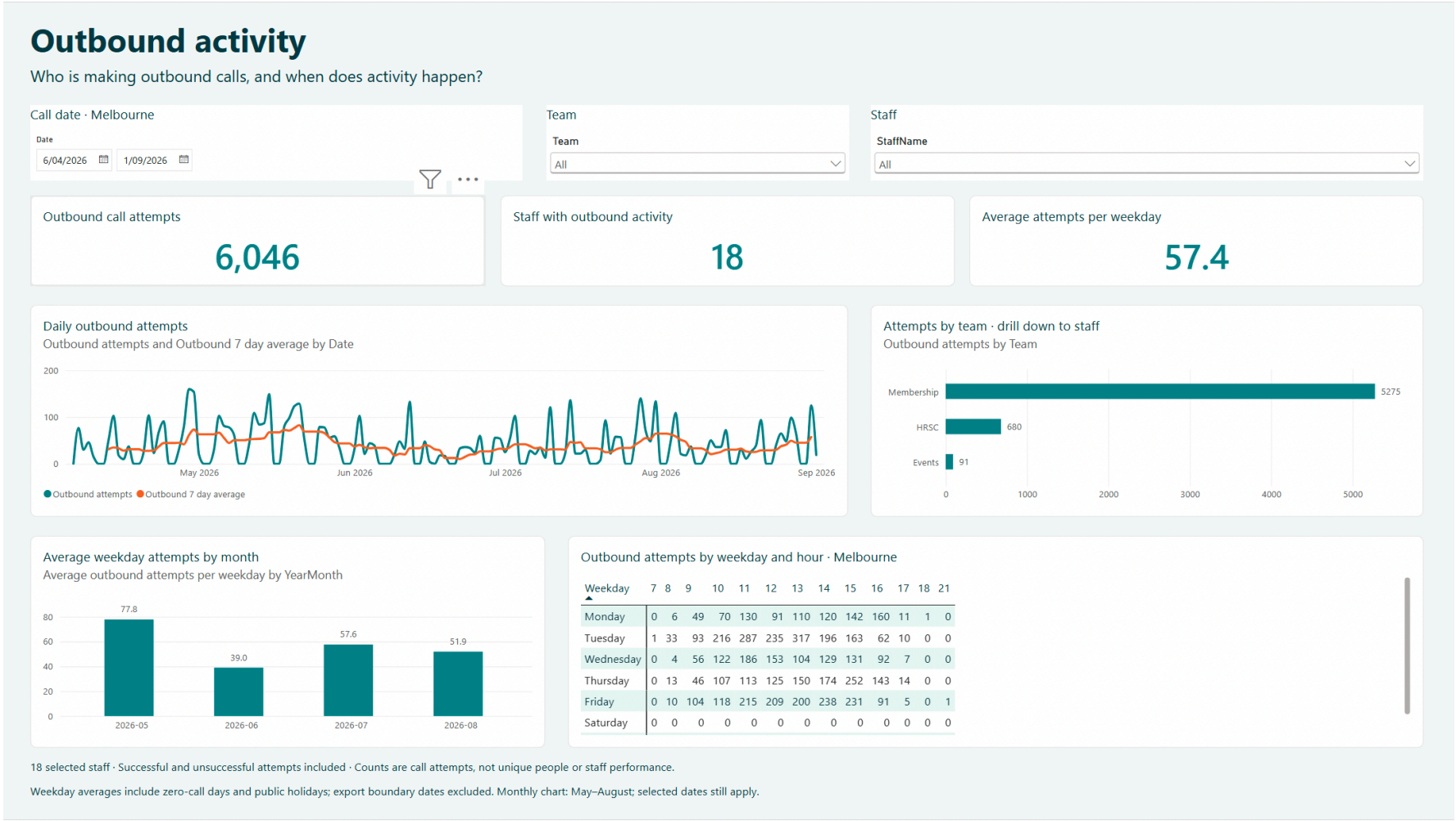


Figure 11: *Outbound activity. Date, team and staff filters support exploration of attempts, while the team chart provides a staff drill-down. Counts describe attempts by the approved accounts; they are not measures of individual productivity.*

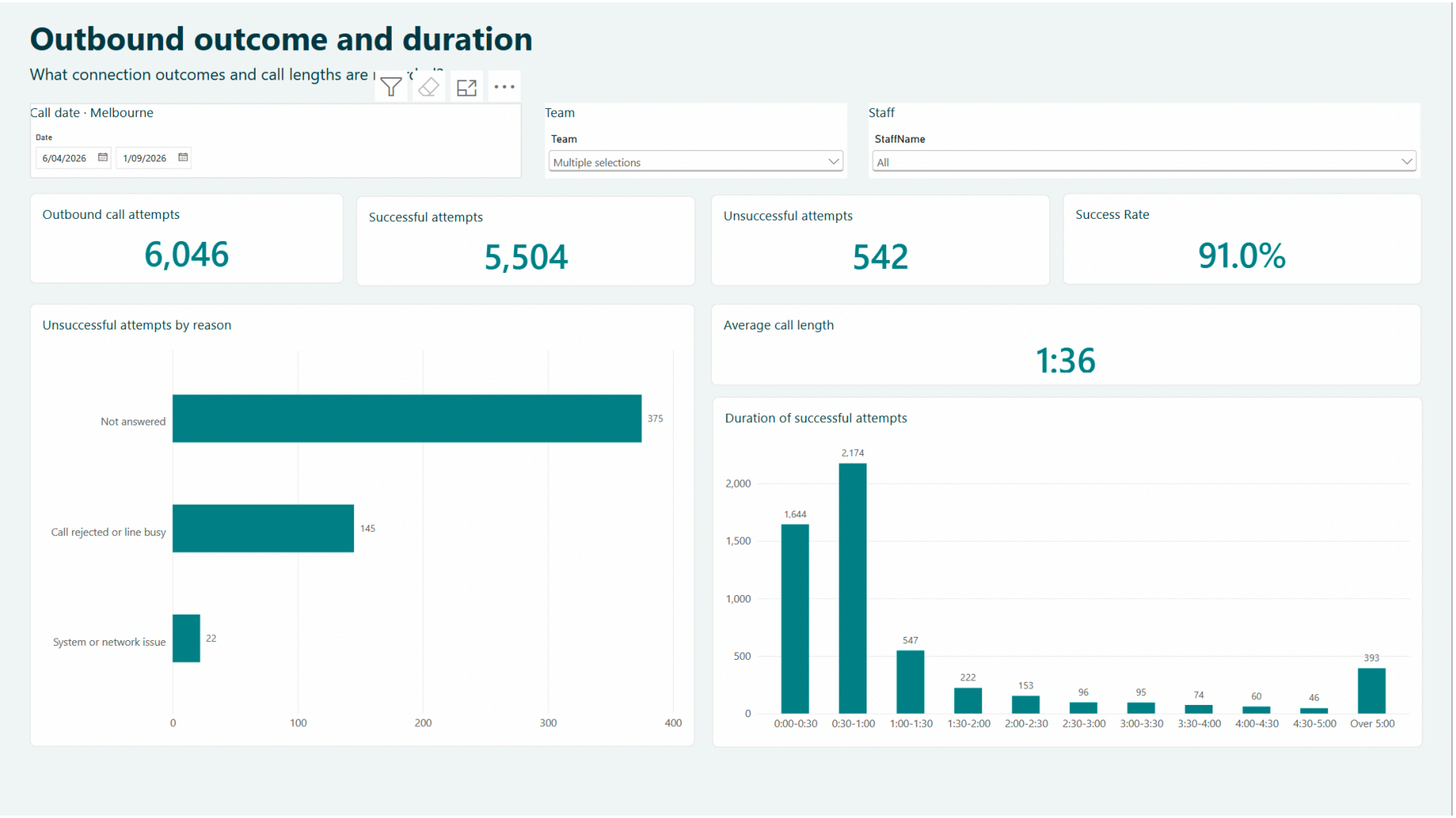


Figure 12: *Outbound outcome and duration. The page combines technical success status with durations of technically successful attempts. The displayed "Not answered" category includes SIP 487, meaning "Request Terminated" (Rosenberg et al., 2002); it should not be interpreted as a confirmed count of unanswered customer calls.*